Practical Submission Template

Machine Learning Course - 316316

📝 STUDENT DETAILS:

Name: Shravya Pravin Mushan

Roll Number: 3264

Date of Submission: 10-01-2026

Practical Number: 1

IDE Used: Jupyter, Anaconda, VS Code, Google Colab, PyCharm

📋 PRACTICAL OVERVIEW:

Practical: Installation of IDE with Necessary Libraries

Duration: 2 Hours

Course Code: 316316 - Machine Learning

Semester: 6th

✅ LEARNING OUTCOMES VERIFICATION:

Check off each learning outcome you have completed:

LO1: Installed and configured Python 3.9+ on system

LO2: Created and activated virtual environment successfully

LO3: Installed all required ML libraries (NumPy, Pandas, Matplotlib)

LO4: Set up and tested Jupyter Notebook environment

LO5: Verified all installations with import statements

LO6: Created requirements.txt file with all dependencies

LO7: Wrote and executed basic Python test programs

LO8: Resolved any installation issues encountered

LO9: Understood IDE functionality and ML development setup

📊 SETUP INFORMATION:

System Information:-

Operating System: Windows

OS Version: 10.0.26200 N/A Build 26200

System Architecture: 64-bit

RAM Available: 4.8 GB

Disk Space Available: 20956536832 GB

Python Installation Details:-

Python Version: 3.13.9

Installation Path: C:\Users\SHRAVYA\AppData\Local\Microsoft\WindowsApps\python.exe

Python Executable: C:\Users\SHRAVYA\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe

pip Version:

pip 25.2 from C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.2544.0_x64__qbz5n2kfra8p0\Lib\site-packages\pip (python 3.13)

📦 LIBRARY INSTALLATION VERIFICATION:

Installed Libraries:

Library	Version	Status	Installation Command
numpy	2.4.0	☐ Installed	python -m pip install numpy
pandas	2.3.3	☐ Installed	python -m pip install pandas
matplotlib	3.10.8	☐ Installed	python -m pip install matplotlib
scikit-learn	1.8.0	☐ Installed	python -m pip install scikit-learn
scipy	1.16.3	☐ Installed	python -m pip install scipy
seaborn	0.13.2		python -m pip install seaborn
jupyter	1.1.1	☐ Installed	python -m pip install jupyter

Total Libraries Successfully Installed: 7 / 7

📊 INSTALLATION EVALUATION LOG:

Step-by-Step Installation Process:-

1. **Python Installation:**

   - **Command used:** Python -version
   - **Output:** 3.13.9
   - **Status:** ☐ Success ☐ Failed ☐ Issues: Success

2. **Library Installation (NumPy):**

   - **Command used:** Python -c “import numpy; print(“ok”)”
   - **Output:** ok
   - **Status:** ☐ Success ☐ Failed ☐ Issues: Success

3. **Library Installation (Pandas):**

   - **Command used:** Python -c “import pandas; print(“ok”)”
   - **Output:** ok
   - **Status:** ☐ Success ☐ Failed ☐ Issues: Success

4. **Library Installation (Matplotlib):**

   - **Command used:** Python -c “import matplotlib; print(“ok”)”
   - **Output:** ok
   - **Status:** ☐ Success ☐ Failed ☐ Issues: Success

5. **Library Installation (Scikit-learn):**

   - **Command used:** Python -c “import sklearn; print(“ok”)”
   - **Output:** ok
   - **Status:** ☐ Success ☐ Failed ☐ Issues: Success

🧪 TEST RESULTS:

Test 1: Python Basic Operations:-

Command: python -V

Output: Python 3.13.9

Result: ✓ PASSED

Test 2: Library Imports:-

Command:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
print("All imports successful")
```

Output: All imports successful

Result: ✓ PASSED

Test 3: NumPy Functionality:-

Command:

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("NumPy array:", arr)
print("Mean:", np.mean(arr))
```

Output:

```
NumPy array: [1 2 3 4 5]
Mean: 3.0
```

Result: ✓ PASSED

Test 4: Pandas Data Manipulation:-

Command Used:

```
import pandas as pd
df = pd.DataFrame({'Name':['A','B','C'],'Marks':[78,85,90]})
print(df)
print(df['Marks'].mean())
```

Output: DataFrame displayed, Average Marks = 84.33

Result: ✓ PASSED

Comments: Pandas operations executed successfully.

Test 5: Matplotlib Visualization:-

Command Used:

```
import matplotlib.pyplot as plt
plt.bar(['A','B','C'], [78,85,90])
plt.show()
```

Output: Bar chart displayed

Result: ✓ PASSED

Comments: Visualization generated correctly.

Test 6: Scikit-learn ML Pipeline:-

Command Used:

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
X,y = load_iris(return_X_y=True)
model = LogisticRegression(max_iter=200).fit(X,y)
print(model.score(X,y))
```

Output: Model accuracy displayed

Result: ✓ PASSED

Comments: ML model trained successfully.

📝 CODE AND EXPLANATIONS:

Code Snippet 1: Virtual Environment:-

Purpose: Create isolated Python environment

```python

**Command used:**

python -m venv ml_env

**Activation command:**

source ml_env/bin/activate (macOS/Linux)

or ml_env\Scripts\activate (Windows)

**Explanation:**

This command creates a virtual environment that isolates Python packages from the system-wide installation. It helps avoid version conflicts between projects and keeps dependencies organized. Activating the environment ensures all installed libraries are used only within this project.

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**Code Snippet 2: Library Installation:-**

**Purpose:** Install all required ML libraries

```python

**Commands used:**

pip install numpy pandas matplotlib scikit-learn scipy seaborn jupyter

**Explanation:**

This command installs essential libraries required for data analysis, visualization, and machine learning. NumPy and SciPy handle numerical computations, Pandas manages data manipulation, Matplotlib and Seaborn are used for visualization, Scikit-learn provides machine learning algorithms, and Jupyter enables interactive notebook execution.

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**Code Snippet 3: Verification Script:-**

**Purpose:** Verify all libraries are installed correctly

```python
import sys
print(f"Python Version: {sys.version}")
try:
    import numpy as np
    print(f"NumPy: {np.__version__} ✓")
except ImportError:
    print("NumPy: FAILED ✗")
[Continue for all libraries...]
```

Explanation:

This script checks whether Python and all required libraries are installed correctly by importing them. If a library imports successfully, its version is displayed; otherwise, an error message is shown. This helps quickly confirm that the environment setup is complete and error-free.

🎯 PRACTICAL COMPLETION CHECKLIST:

- Python 3.9+ installed
- Virtual environment created and activated
- All libraries installed (NumPy, Pandas, Matplotlib, Scikit-learn, etc.)
- All imports work without errors
- At least 5 tests executed and passed
- Jupyter Notebook (or chosen IDE) working
- Submission files organized

📌 DECLARATION:

I hereby declare that:-

1. ☐ I have completed this practical on my own
2. ☐ The code and explanations are my original work
3. ☐ I have not copied from other students
4. ☐ I have used reference materials and documented them
5. ☐ All tests have been executed on my system
6. ☐ My submission is complete and ready for evaluation

Student Signature: Shravya P.M.

Date: 10-01-2026

👩‍🏫 EVALUATOR COMMENTS:

To be filled by instructor/evaluator:-

- **Technical Competency:** ___ / 10
- **Python Installation:** ___ / 2
- **Virtual Environment:** ___ / 2
- **Library Installation:** ___ / 2
- **Testing & Verification:** ___ / 2
- **Documentation:** ___ / 2
- **Understanding & Application:** ___ / 10
- **Concept Understanding:** ___ / 3
- **Problem-Solving:** ___ / 3
- **Code Explanation:** ___ / 2
- **Troubleshooting:** ___ / 2
- **Presentation & Organization:** ___ / 5
- **Documentation Quality:** ___ / 2
- **Folder Structure:** ___ / 1
- **Submission Organization:** ___ / 2
- **Overall Performance:** ___ / 25

Evaluator Comments:

_____

_____

_____

Strengths:

_____

Areas for Improvement:

_____

Suggestions for Next Practical:

_____

Evaluator Name: _____

Evaluator Signature: _____

Date of Evaluation: _____

📞 REFERENCE LINKS:

Useful resources for this practical:-

- **Python Installation:** https://www.python.org/downloads/
- **Virtual Environments:** https://docs.python.org/3/tutorial/venv.html
- **NumPy Docs:** https://numpy.org/doc/
- **Pandas Docs:** https://pandas.pydata.org/docs/
- **Matplotlib Guide:** https://matplotlib.org/
- **Scikit-learn Docs:** https://scikit-learn.org/
- **Jupyter Notebook:** https://jupyter.org/
- **Stack Overflow:** https://stackoverflow.com/

Version: 1.0

Last Updated: December 2025

For Course: 316316 - Machine Learning

Practical: 1 of 15

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